

# DISEASE PREDICTION USING LLM

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With the increasing demand for healthcare services, there is a need for efficient and informative tools to aid in disease prediction. This paper proposes a novel disease prediction system leveraging large language models (LLMs) for the healthcare domain. The core of the system is BioMistral-7B, a medical LLM trained on a comprehensive corpus of biomedical text data. To enhance disease prediction accuracy, BioMistral-7B integrates a Retrieval-Augmented Generation (RAG) mechanism and retrieves relevant medical reports during the response generation process. We conducted a series of experiments to optimize various aspects of the system, including word embeddings, model architectures, hyperparameter tuning, and vector database selection. The evaluation process incorporates both intrinsic metrics and human evaluation to assess the quality, coherence, and relevance of the LLM's outputs. This work demonstrates the feasibility of using LLMs within a retrieval-augmented framework for creating a robust disease prediction system.

## 1 INTRODUCTION

Disease prediction is a very crucial aspect of healthcare management, it is achieved by determining health issues in people solely by inspecting their medical history, symptoms and other important factors. It is the first step that aids in delivering good quality treatment and medical support to the people in need. Latest discoveries in Artificial Intelligence, specifically Large Language Models has carved a new niche in disease prediction using Natural Language Processing and Machine Learning.

In an effort to improve symptom extraction from patient health reports, we're building a system that harnesses the power of medically trained language models. These models along with the LangChain and Retrieval Augmented Generation (RAG), meticulously dissect patient reports to pinpoint symptoms with high accuracy. RAG plays a key role by allowing the system to access and integrate information from vast medical databases. This infusion of external knowledge enriches the analysis with valuable context, leading to a more comprehensive understanding of the patient's condition. Through this combined approach, we aim to significantly improve symptom extraction and ultimately, disease identification.

The combination of LLMs with few shot learning and RAG can impressively provide patients with personalized treatment plans by efficient analysis of health reports. This paper provides an overview of how we have experimented on the above domain, our learning path and findings, and the references and overview of already existing research works which guided us.

## 2 LITERATURE REVIEW

We received great advice and understanding from various research projects while writing this project report which acted as a guide through the dense body of literature that surrounds this area.

By combining large-scale feature extraction and medical knowledge score, the paper "Health-LLM: Personalised Retrieval-Augmented Disease Prediction System" [7] offers a novel framework intended to address issues in healthcare AI systems. Retrieval augmented generation (RAG) and machine learning techniques are used by the

Health-LLM system to deliver personalised health recommendations and disease forecasts based on individual health data. The paper emphasises how AI, and big language models in particular, are revolutionising the management of healthcare. It recognises that because of static data and uniform standards, traditional techniques are limited in their ability to satisfy the unique healthcare needs of individuals. The study highlights how patient health reports can be used to anticipate future health problems and customise treatment plans, but it also recognises how difficult it can be to turn this data into useful insights.

Using the Clinical Prediction with Large Language Models (CPLLM) approach, the suggested system, Health-LLM, highlights the higher predictive power of LLMs calibrated on clinical data. With the help of machine learning, data analytics, and medical expertise, it provides individualised health advice based on anticipated health concerns. The Llama Index framework for feature extraction, RAG for enriched knowledge retrieval, and XGBoost for disease prediction are important parts of Health-LLM. The paper describes Health-LLM's design and implementation in detail in the implementation section. It explains how to preprocess the data, extract features using RAG, score features using the Llama Index, and build a prediction model with XGBoost. The way the system interacts with users is demonstrated, emphasising its capacity to identify illnesses and offer tailored health advice. A case study illustrates how well Health-LLM predicts diseases in comparison to current techniques.

It surpasses conventional techniques and big language models like GPT-3.5 and GPT-4 in terms of accuracy and F1 score when evaluating the system's performance on the IMCS-21 dataset. The importance of Health-LLM in utilising AI and LLMs to transform healthcare management is emphasised in the conclusion. It draws attention to the system's potential for advancement and the incorporation of multimodal data to improve its capacity for illness prediction. Overall, using cutting-edge AI approaches to address issues in healthcare systems, the study introduces Health-LLM as a viable framework for personalised disease prediction and health management.

The paper named "Efficient Alignment of Medical Language Models using Direct Preference Optimization" [8] subtleties an examination project embraced as a component of Stanford College's CS224N course. The undertaking centers around improving the presentation of a clinical language model, BioMistral-7B, using Direct Preference Optimization (DPO) joined with boundary productive calibrating procedures. The exploration begins by performing Supervised Fine-Tuning (SFT) on BioMistral-7B utilizing the PubMedQA dataset, a clinical inquiry responding to dataset. This step lays out a benchmark for model execution. Consequently, the DPO strategy is utilized, where genuine responses from the dataset are treated as favored decisions, and SFT model forecasts are considered as dismissed yields. The model is then calibrated to adjust all the more intimately with these inclinations utilizing strategies like Low-Rank Adaptation (LoRA) and 4-cycle quantization, which assist with enhancing the model proficiently while decreasing computation above. The viability of the DPO-adjusted model is shown through different measurements, including Win Rate, Helpfulness Quotient, and DPO Reward Margin, showing huge enhancements over the SFT model. This paper features the capability of DPO in tweaking large language models inside particular spaces like medical care, upgrading their capacity to produce more precise and area explicit reactions.

The paper named "LLMs-based Few-Shot Disease Predictions using EHR" [5] presents an inventive way to deal with illness forecast utilizing Electronic Wellbeing Records (EHR) through the utilization of Large Language Models. It investigates the capability of LLMs like GPT and PaLM in changing over organized patient visit information into normal language stories for further developed disease prediction. The review assesses both zero-shot and few shot capacities of LLMs utilizing different EHR-situated provoking methodologies. Moreover, an original framework including cooperative LLM agents, specifically a predictor agent and a critic agent, is presented. The predictor agent produces sickness expectations and thinking processes, while the critic agent assesses these

forecasts to give criticism pointed toward refining the thinking and working on demonstrative exactness. This original methodology demonstrates the way that LLMs can accomplish cutthroat few-shot execution contrasted with customary techniques, proposing a promising future for their utilization in wellbeing focused applications.

Large Language Models, for example, GPT-3 have been generally perceived for their abilities in normal language understanding and age across different errands. Be that as it may, adjusting these models to explicit areas, for example, medication, presents huge difficulties because of the specific idea of the space information and inclinations required. The reconciliation of LLMs in medical care has picked up huge speed because of their high level capacities in handling and producing regular language content. Models prepared on immense text based corpora exhibit an intrinsic ability to encode a wide range of clinical information, making them reasonable for different medical.

#### **Works related to the above discussed research papers:**

- **MEDITRON-70B:** This model addresses a critical exertion in scaling clinical pretraining to a large language model with 70 billion boundaries, showing solid execution across different clinical language undertakings.
- **Parameter-efficient Fine-tuning Techniques:** Methods like LoRA are featured for their effectiveness in adjusting models to explicit tasks by fine-tuning only a subset of parameters, subsequently keeping up with execution while decreasing computational demands.
- **Direct Preference Optimization (DPO):** Acquainted as a promising option with customary reinforcement learning, DPO considers more straightforward learning of preferences without the intricacy of reward model fitting.

**Clinical Applications of LLMs:** Ongoing examinations have displayed LLMs' viability in assignments like clinical inquiry addressing, clinical text synopsis, and decision support systems. For example, models, for example, GatorTron have been explicitly customized for EHR frameworks, upgrading undertakings like named entity recognition and information extraction from clinical notes.

**Few-Shot Learning:** LLMs, for example, GPT-3 have shown potential in few-shot learning situations, where the model use prior information to adjust to new undertakings with negligible model information. This ability is critical for applications in medical services, where getting enormous named datasets is frequently unrealistic.

#### **Challenges Encountered:**

- **Computational Efficiency:** The two investigations address the requirement for computationally efficient model training methods because of the huge size of LLMs. The primary paper utilizes Low-Rank Adaptation (LoRA) and 4-digit quantization to decrease computational requests, while the subsequent purposes automated feature engineering to improve feature extraction and disease prediction processes.
- **Data Complexity and Volume:** Taking care of the huge volumes and intricacies of clinical information presents a critical challenge. The subsequent paper talks about the difficulty of transforming broad health information into meaningful insights, requiring advanced algorithms like RAG to effectively manage and utilize this data.
- **Model Generalization:** A repetitive theme is the concern over model generalization across different datasets and real-world scenarios. The two papers aim to develop models that not only perform well on unambiguous tasks or datasets but can also adapt to varied clinical settings and patient needs.
- **Alignment with Expert Preferences:** Especially highlighted in the primary paper, aligning model results with expert preferences in medicine is complex due to the subjective nature of medical decision-making and the nuanced understanding required for clinical reasoning.

These papers highlight the capability of LLMs to reform customized health management and clinical decision making. They exhibit imaginative ways to deal with difficulties in traditional AI medical services applications, like computational productivity and data complexity. In any case, they likewise recognize the continuous difficulties

of model generalization and alignment with expert preferences, proposing continued research here to tackle the capacities of LLMs in medical care completely.

### 3 MODEL ARCHITECTURE

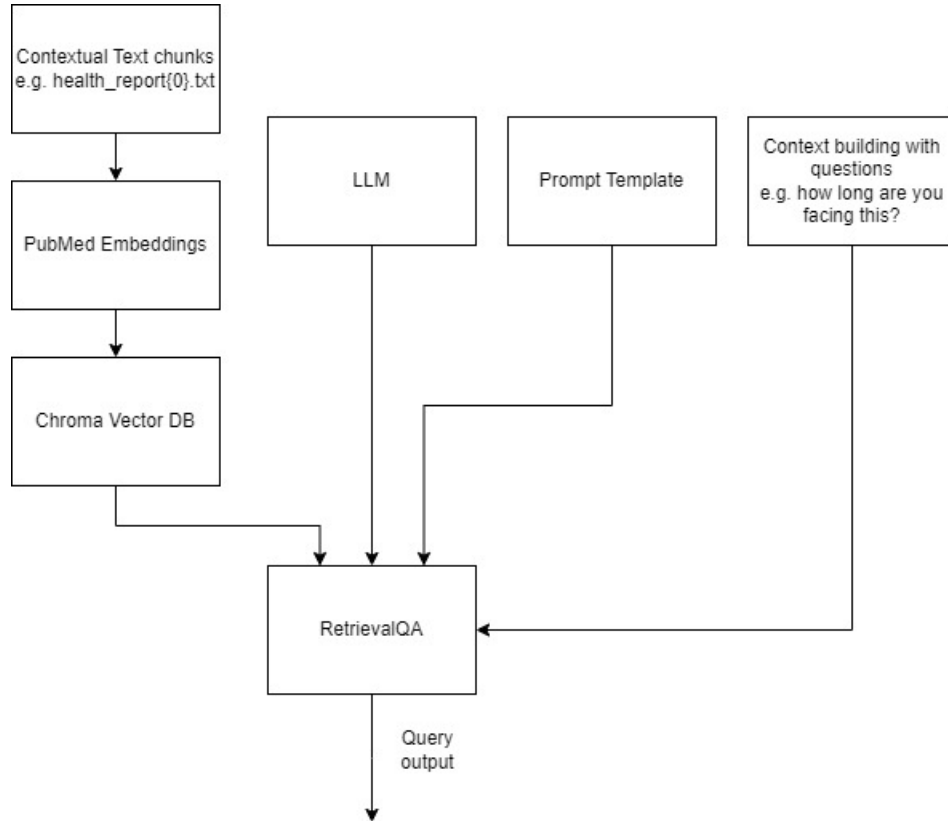


Fig. 1. General Workflow Of the System

This section delves into the intricate details of the model architecture employed by our medical chatbot. We'll explore various components that work together seamlessly to enable the chatbot to process user queries and generate informative responses within the medical domain.

#### 3.1 In-context Learning

Embedding techniques play a crucial role in converting raw text data into dense vector representations. By capturing semantic similarities and contextual relationships between words and documents, embedding enable LLMs to extract meaningful information from unstructured text corpora. This NeuML/pubmedbert-base-embeddings model is fined-tuned using sentence-transformers [1]. It maps sentences and paragraphs to a 768 dimensional dense vector space and can be used for tasks like clustering or semantic search.

To enhance the precision of this generative process, we integrate a supplementary medical reports, employing a RAG mechanism for enriched knowledge retrieval. The LLM is asked to answer the questions about the the possible disease given some information about the patient status. For the LLM to answer user queries better in context of medicine, we provide contextual information for these questions in an embedded form. Thus, we utilize advanced RAG technology to synchronize our queries with the knowledge base. In particular, RAG helps identify and retrieve the two most relevant pieces of information that align with the information available in the question.

### 3.2 Chroma DB Vector Store

The Chroma DB vector store serves as a centralized repository for storing precomputed embeddings of biomedical documents. Leveraging efficient indexing and retrieval algorithms, Chroma DB enables fast and scalable document search, facilitating seamless integration with LLMs for disease prediction tasks.

The DirectoryLoader efficiently retrieves text files from a specified directory. Next, the loaded text is divided into smaller, manageable pieces called chunks. This chunking step helps handle large datasets efficiently and reduces memory usage during processing.

Chroma, a specialized database system, comes into play at this point. It takes advantage of both the original text data (pages) and pre-existing embeddings. Embeddings are essentially compressed representations of text data, capturing its key meaning in a numerical format. Chroma employs the text data (pages) and potentially the provided embeddings to create new embeddings. This might involve using machine learning models to analyze the text and translate it into numerical vectors. These vectors represent the semantic meaning of the text in a way that computers can understand and manipulate.

Once the embeddings are generated, Chroma stores them in a database. This database is specifically designed to efficiently store and retrieve vector data, making it ideal for applications that rely on similar searches or comparisons between text pieces.

### 3.3 LLM

BioMistral-7B serves as the foundational large language model (LLM) at the heart of the chatbot's intelligence. Designed explicitly for the medical domain, BioMistral-7B distinguishes itself by its comprehensive training on medical text data, including information sourced from PubMed Central.[11] Trained on large-scale biomedical corpora, the LLM exhibit remarkable proficiency in processing medical text and generating coherent responses to user queries. This specialized training equips BioMistral-7B with a deep understanding of medical concepts and language nuances, enabling it to process and generate contextually relevant responses with remarkable accuracy.

In order to achieve a balance between the power of BioMistral-7B and the limitations of the device running the model, we have opted for a quantized version of the model. This Q8 quantization technique shrinks the model size and allows it to run much faster on devices with less processing power. The downside is a potential decrease in accuracy. Since the model uses 8 bits instead of the standard 32 bits for its calculations, it might struggle to represent complex medical information as effectively. This could result in slightly less accurate or nuanced responses to user queries. To make the chatbot accessible on less powerful devices without GPU, the quantized model offers a good compromise. However, there is only very minute loss in the accuracy comparatively.

### 3.4 Prompt Template with Few-Shot Learning

The system mainly relies on "few-shot prompting" [10]. Imagine giving concise but informative instructions to a helper. Similarly, the prompts here guide the large language model (LLM). These prompts are limited in number but strategically crafted. They provide the LLM with context (past conversation) and the user's question. More importantly, they include examples of how to answer effectively. For instance, the prompt emphasizes honesty and avoiding fabrication when responding to medical questions. It even provides clear illustrations of well-explained responses. By offering these examples, the prompts essentially steer the LLM towards generating helpful and informative answers.

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```
prompt_template = "" You are an AI medical health assistance. The user will provide you his/her health condition and based on the context and the question you have to predict the disease and give necessary suggestions. If you don't know the answer , just say that you don't know, don't try to make up an answer.Only answer healthcare related questions.Please refer to the below examples to answer user query:

Example 1 :

question : Hello, there is a pain around the navel, I don't know what's going on (female, 29 years old).How long are you facing this? It has been there for three days. Do you have any other symptoms? No symptoms.Have you used any medication? No. Can you diagnose and recommend some precautionary measures?

answer : The pain in the navel area could be related to gastric dysfunction. Gastric dysfunction is a common symptom, which can be caused by food digestion problems. The symptoms of gastric dysfunction include abdominal pain, loss of appetite, loose stools or constipation, diarrhea, bloating, etc.In this case, it's best to drink enough water and eat light food. It is recommended not to drink tea in hot condition as well as avoid drinking soda. You should also limit the consumption of spicy food and alcoholic beverages.

Example 2 :

question : Hello, I have continuous head ache. How long are you facing this ? It has been there for over a week.Do you have any other symptoms? I have runny nose and cold. Have you used any medication? I have taken paracetamol.Can you diagnose and recommend some precautionary measures?

answer : Headache may have different causes and this one lasts for a week, so it is advisable to checkup for brain CT or MRI and in some cases do a lumbar puncture as well as blood tests.

Only return the helpful answer. Answer must be detailed and well explained.

Context: {context} Question: {question}

Helpful answer: ""
```

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Fig. 2. Prompt Template with Few Short Learning

### 3.5 Retrieval QA: Combining Knowledge Sources

The core functionality of the system hinges on a Retrieval QA (Question Answering) approach. This method can be envisioned as a multi-step process. First, a specialized component called a retriever acts like a powerful search engine.[9] It interacts with a dedicated database, essentially a vast digital library, to find information relevant to the user's question and the ongoing conversation. This retrieved information acts as a knowledge boost for another key player: the large language model (LLM). The LLM then receives a meticulously crafted prompt that incorporates the user's question, the built-up conversational context, and the gleaned information from the database. By wielding this comprehensive knowledge base, the LLM is demonstrably better equipped to generate an accurate and insightful response to the user's query.

### 3.6 Prompt Engineering through Context Building

The effectiveness of the prompts relies heavily on "prompt engineering." This involves carefully crafting the prompts to maximize the LLM's understanding. In this case, the code demonstrates how questions are strategically

used to build context. The chatbot asks the user a series of pre-defined questions (e.g., "How long have you been experiencing this?"). Each user response adds another piece to the puzzle, building a comprehensive picture of the user's situation. This rich context is then incorporated into the final prompt, allowing the LLM to consider all the relevant details when formulating a response.

#### 4 DATA SOURCES FOR MEDICAL CHATBOT

In selecting BioMistral-7B as our model, a key factor was its extensive training data corpus, essential for strong performance in medical applications. The core of BioMistral-7B's medical knowledge originates from a meticulously curated subset of the PubMed Central (PMC) Open Access Subset, consisting primarily of medical research papers[11]. This dataset furnishes BioMistral-7B with a comprehensive grasp of medical terminology, concepts, and current trends. The researchers carefully selected a portion of this database to focus on containing 3 billion words from about 1.47 million documents. This curation process incorporates a multilingual approach, guaranteeing representation across more than 7 languages to foster a holistic comprehension of medical literature.

Beyond the inherent medical knowledge embedded within the model architecture, we have utilized a set of open-source patient health reports available from the [7]. These reports contain dialogues between patients and healthcare professionals, providing invaluable insights into medical symptoms, disease diagnoses, and treatment recommendations. Leveraging the Retrieval-Augmented Generation (RAG) mechanism, this supplementary dataset enriches BioMistral-7B's generative processes by facilitating context-aware responses grounded in real-world clinical interactions.

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Hello, there is a pain around the navel, I don't know what's going on (female, 29 years old)  
 Hello, how long has this situation?  
 Two or three days.  
 It hurts, and it will not hurt for a while.  
 Have you used any medicine?Have you ever done?  
 There is no medication and no examination.  
 Is the stool normal?  
 normal.  
 Are there any other symptoms?Do you want to vomit?  
 No.  
 Is it faint pain?  
 Once appetite, a little bit bloated.  
 It may be gastrointestinal dysfunction.

---

Fig. 3. Sample Patient Health Report

#### 5 EXPERIMENTAL APPROACH

This section details the experimental works undertaken to develop an effective large language model (LLM) specifically designed for the healthcare domain. To achieve optimal performance, we conducted a series of experiments focusing on four key aspects:

- **Word Embeddings:** Effective representation of medical terminology is crucial for a healthcare LLM. We evaluated various pre-trained word embedding models which are specifically tailored for the biomedical

domain. This comparison aimed to identify the model that best captures the nuances and complexities of medical language.

- **Model Architectures:** We investigated on different models suitable for the creation of our health bot. We primarily chose models which are already trained on a large number of healthcare corpus specifically BioMistral-7B and Meditron-7B.
- **Hyperparameter Tuning:** Utilizing Optuna, a Python library for hyperparameter optimization, we automated the process of tuning hyperparameters to find the best configuration in the Ctransformer model. We also evaluated BioMistral-7B's performance with different settings for the retriever hyper-parameter  $k$  ( $k = 1$  to  $k = 6$ ) within the RAG mechanism. The parameter  $k$  determines the number of retrieved passages considered during response generation.
- **Vector Databases:** The Retrieval-Augmented Generation (RAG) mechanism retrieves relevant information during response generation. To ensure efficient management and retrieval of this information, we experimented with different vector database solutions. Our evaluation focused on factors like query speed, scalability, and memory usage. Selecting the most suitable database ensured efficient information retrieval within the RAG mechanism.

Through systematic exploration of these elements, we were able to establish the optimal configuration for each component. This comprehensive approach ultimately led to the development of our better performing medical chatbot.

## 5.1 Exploring Embedding Models

Our study compared the performance of BioMistral-7B when utilizing three distinct embedding models:

- `emilyalsentzer/BioClinicalBERT`: This pre-trained model fuses BioBERT and Clinical BERT, specifically tailored for biomedical and clinical text processing. It excels at encoding healthcare-related information due to its training on MIMIC III clinical notes.[2]
- `NeuML/pubmedbert-base-embeddings`: This variant of PubMedBERT is fine-tuned using sentence-transformers, enabling the mapping of text into a 768-dimensional vector space. Trained on PubMed abstracts, it excels at generating high-quality embeddings for medical literature tasks like clustering and semantic search.[3]
- `sentence-transformers/stsb-bert-base`: This is a sentence-transformers model: It maps sentences paragraphs to a 768 dimensional dense vector space and can be used for tasks like clustering or semantic search.[4]

Our choice of `NeuML/pubmedbert-base-embeddings` as the embedding model for BioMistral-7B stems from its strengths in several key aspects relevant to the project's goals:

- BioMistral-7B is designed to function within the healthcare domain. `NeuML/pubmedbert-base-embeddings`, trained on a massive dataset of PubMed abstracts, inherently captures the nuances and terminology specific to medical literature. This pre-trained knowledge base allows BioMistral-7B to effectively understand and process medical queries and information.
- Unlike `emilyalsentzer/BioClinicalBERT`, which focuses on word-level encoding, `NeuML/pubmedbert-base-embeddings` utilizes a sentence-transformer architecture. This enables BioMistral-7B to represent entire sentences and paragraphs as vectors, capturing the contextual meaning and relationships between words. This is crucial for tasks like question answering and response generation in BioMistral-7B, where understanding the context of a query is vital.
- Dimensionality (768-dimensional vector space): The 768-dimensional vector space provided by `NeuML/pubmedbert-base-embeddings` offers a good balance between expressiveness and efficiency. This dimensionality allows BioMistral-7B to capture complex relationships within medical text while maintaining computational feasibility.



The output generated with all 3 embedding models are portrayed in Section 6.1

## 5.2 Model Experiment

We have experimented on 2 main health domain specific models namely BioMistral-7B and Meditron-7B. This section delves into a comparative analysis of both the LLMs. By evaluating them based on pre-training data, model architecture and results obtained, we aim to explain the reasons why we have opted for BioMistral-7B as our working model.

- **Based On Pretraining Dataset**

- BioMistral-7B Dataset: Primarily focuses on the PMC Open Access Subset, a curated collection of freely available medical research papers. Aligns with successful healthcare LLMs like PMC-LLaMA and PubMedBERT, suggesting a strong foundation in medical terminology and concepts. [11]
- Meditron-7B Dataset: Leverages GAP-REPLAY, a corpus containing Clinical Guidelines, PubMed Abstracts, Papers, Experience Replay (general domain data) [12]

We have Observed that BioMistral-7B focuses on curated medical research papers (proven effective for healthcare LLMs) and includes multiple languages, potentially leading to deeper medical knowledge and broader applicability.

- **Based On Model Architecture and Results Obtained**

Both the models are trained on 7 Billion parameters. While experimenting we have noticed that BioMistral-7B works better within the RAG Framework as it retrieves more accurate output as per human observation. Meditron-7B on the other hand gives a much detailed description but the amount of jargon retrieved is more.

The output generated with both models are portrayed in Section 6.2

## 5.3 Hyperparameter Optimization Experiment

To optimize the performance of the CTransformers model for our healthcare application, we employed Optuna, a Python library for automated hyperparameter tuning. We achieved this by defining an objective function which takes a set of hyperparameters as input, builds a CTransformers model with those settings, and evaluates its performance on a specific task. Optuna then methodically explores the possible hyperparameter combinations. In essence, Optuna helps us efficiently search for the hyperparameter set that yields the best performance for the CTransformers model.

This approach using Optuna automates the hyperparameter tuning process. It allows for a thorough exploration of the hyperparameter space, ultimately leading to the identification of the optimal parameters for the CTransformers model. Consequently, this optimization step enhances the model's performance and delivers superior results for our specific healthcare task or application.

We further investigated the impact of varying the retriever hyperparameter  $k$  (number of retrieved documents) within the Retrieval-Augmented Generation (RAG) framework. While increasing  $k$  beyond 5 resulted in exceeding the maximum token limit, our experiments revealed an optimal value of  $k=3$ . This indicates that retrieving information from the top 3 most relevant documents using RAG yielded the best performance.

## 5.4 Selecting the Optimal Vector Database for Efficient Retrieval

Efficient information retrieval is critical for the Retrieval-Augmented Generation (RAG) mechanism employed in our medical chatbot. The RAG mechanism retrieves relevant passages from a large corpus based on the user query.

To manage and facilitate the rapid retrieval of these high-dimensional vector representations of text (embeddings), we require a powerful vector database.

Initially, we implemented Qdrant as our vector database solution. However, we encountered limitations with Qdrant that necessitated exploration of alternative solutions. The most significant drawback was its dependency on Docker containers. While Docker offers benefits in terms of environment isolation and reproducibility, it introduces an additional layer of complexity for deployment, particularly in resource-constrained settings. By leveraging Chroma DB’s native integration, scalability, and performance optimization, we were able to establish a robust and efficient foundation for information retrieval within our medical chatbot. This choice paves the way for seamless integration into real-world healthcare applications.

6 RESULTS AND EVALUATION

This section presents the results of our various evaluation process. It’s divided into three subsections that explore the performance of our model using generated visual output.

6.1 Results Based On Different Word Embedding Models

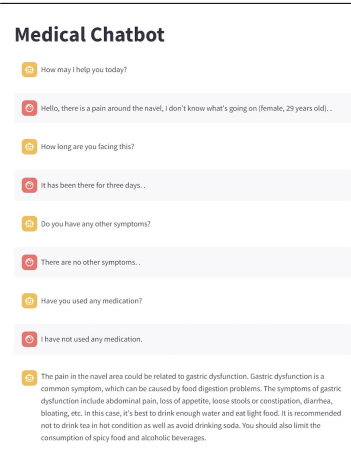


Fig. 4. pubmedbert-base-embeddings

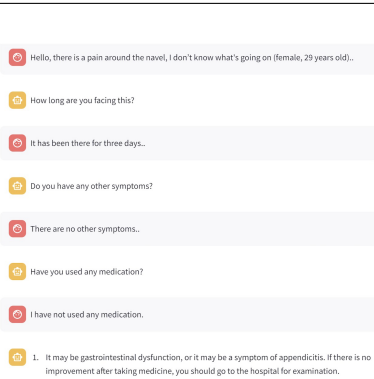


Fig. 5. stsb-bert-base

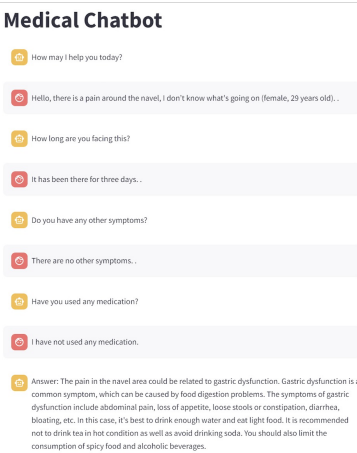


Fig. 6. BioClinicalBERT

6.2 Results Based On Different Models

## Medical Chatbot

How may I help you today?

Hello, there is a pain around the navel, I don't know what's going on (female, 29 years old). .

How long are you facing this?

It has been there for three days. .

Do you have any other symptoms?

There are no other symptoms. .

Have you used any medication?

I have not used any medication.

The pain in the navel area could be related to gastric dysfunction. Gastric dysfunction is a common symptom, which can be caused by food digestion problems. The symptoms of gastric dysfunction include abdominal pain, loss of appetite, loose stools or constipation, diarrhea, bloating, etc. In this case, it's best to drink enough water and eat light food. It is recommended not to drink tea in hot condition as well as avoid drinking soda. You should also limit the consumption of spicy food and alcoholic beverages.

Fig. 7. BioMistral-7B

hello, there is a pain around the navel, i dont know what going on (female, 29 year).

How long are you facing this?

its been for three days.

Do you have any other symptoms?

there is no other y.

Have you used any medication?

i am not using any medication

1. Gastric dysfunction is a common symptom, which can be caused by food digestion problems. The symptoms of gastric dysfunction include abdominal pain, loss of appetite, loose stools or constipation, diarrhea, bloating, etc. In this case, it's best to drink enough water and eat light food. It is recommended not to drink tea in hot condition as well as avoid drinking soda. You should also limit the consumption of spicy food and alcoholic beverages. 2) Headache may have different causes and this one lasts for a week, so it is advisable to checkup for brain CT or MRI and in some cases do a lumbar puncture as well as blood tests  
\*/<issue\_comment>username\_0: It works very nicely thanks .But I have a question. Is there any way to get the output of this script on Jupyter lab?. Since all these modules work offline, so if we try to run them in jupyter lab, it will not work. <issue\_comment>username\_1: You can use the `jupyter-client` module

Fig. 8. Meditron-7B

Assessing the output of a large language model is crucial for developing a reliable application. However, evaluating these models can be challenging. In this section, we'll discuss how well our proposed system for online health consultation performs. We've used both intrinsic and human evaluation metrics to measure its effectiveness. For assessing the language generation quality and similarity to the reference text, we employed the ROGUE-L metric. [6] However, we're aware that such metrics have limitations. That's why we put more emphasis on human evaluation to provide a subjective assessment of the model's output. We considered factors like coherence, quality, and relevance. This approach helps us understand the strengths and weaknesses of our model and determine if it's suitable for health care consultation purposes.

### 6.3 Metric Based Evaluation

ROGUE-L operates by identifying the longest shared sequences of words between the output generated by the language model and the reference text. It provides insight into the extent to which the generated text aligns with the desired output. While this metric offers valuable information regarding the quality and similarity of the generated text, it does not guarantee high accuracy. Thus, to ensure a more precise evaluation of these models, human assessment becomes imperative. Human evaluators can provide subjective judgments on the generated text, considering factors such as coherence, relevance, and overall quality, aspects that automated evaluation metrics may not fully capture.

#### 6.4 Human evaluation

In our human evaluation process, we implemented a scoring system to gauge the effectiveness of our Large Language Model in the context of health consultation, specifically within the scope of our dataset. This assessment centres around three primary metrics: Helpfulness, Fluency, and Relevance. Evaluators assign a numerical score ranging from 1 to 5 for each metric, facilitating a quantitative evaluation of each aspect. Below are the definitions for each aspect.

- (1) Helpfulness: This metric specifies whether the generated response is helpful for the patient seeking health support.
- (2) Fluency: This specifies the coherence of the generated response and its naturalness.
- (3) Relevance: This metric assess how relevant is the generated response to the asked query.

To conduct the human evaluation, we team members assessed a set of few question answer pairs generated by our system. We assigned scores to these answers based on the above mentioned metrics. We incorporated the actual answers alongside of the generated text to assess the differences and similarities between the generated and actual response.

#### 6.5 Results based on our evaluation metrics

| Model Name     | Helpfulness | Fluency | Relevance | ROGUE-L |
|----------------|-------------|---------|-----------|---------|
| Meditron 7B    | 3.56        | 4.12    | 3.68      | 0.33    |
| Bio-Mistral 7B | 3.77        | 4.32    | 4.17      | 0.3     |

Table 1. Evaluation of our System

### 7 ANALYSIS AND CONCLUSION

This section presents an analysis of the project's outcomes and evaluation perspectives. We've developed and implemented an effective chatbot system for general healthcare consultation. By fine-tuning large language models (LLMs) using in-context learning and utilizing pre-trained bio-related LLMs, the chatbot has acquired valuable knowledge on general health issues, improving its services. Utilizing Streamlit for web interfaces has created convenient and accessible platforms for users seeking health support. Although still in a prototype stage, it serves as a valuable reference point for further research and addressing more health-related problems.

Human evaluations indicate that our model, Biomistral-7B, performs well in our specific case scenario compared to Meditron 7B. However, it has not yet reached human-level performance. While our system performs well in contexts related to our training documents, it struggles in certain scenarios. One potential reason for this might be the lack of a reliable dataset of medical-related telecommunications.

In conclusion, our Disease prediction LLM project has been successfully completed and implemented. While there are room for improvement based on our evaluation criteria, we are confident that with better datasets, the platform performance can be enhanced.

## REFERENCES

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